

Mission-Grade TT&C Communication System for CubeSats

System Design Report

SATC-6 · Task IV · Document 1 of 4

1. Introduction and Scope

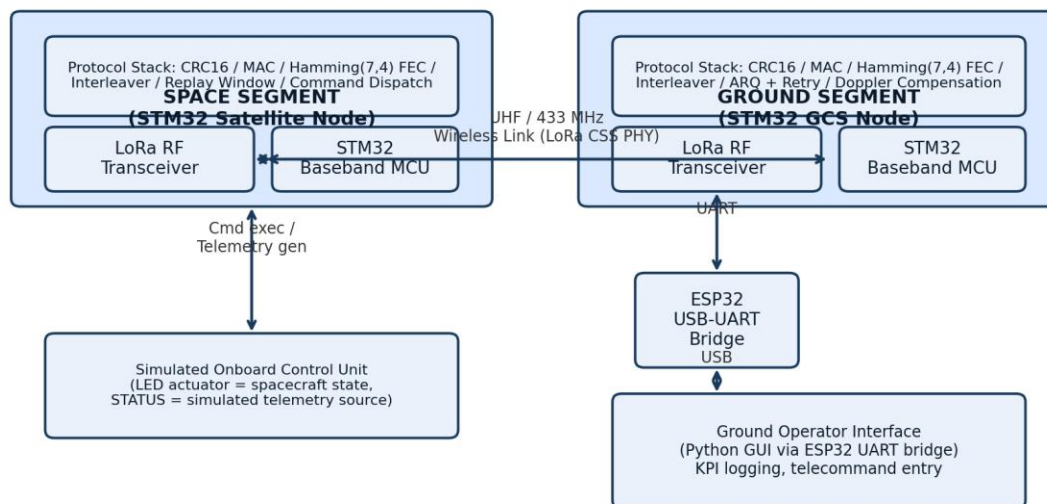
This report presents the system-level architecture and engineering rationale for the Telemetry, Tracking, and Command (TT&C) communication subsystem developed for Task IV of the SATC-6 competition. It establishes traceability between the mandatory functional and performance requirements defined in the competition call and the design decisions made for both the space segment and the ground segment.

The subsystem targets a Low Earth Orbit (LEO) CubeSat mission operating in the 500-700 km altitude band, using the UHF band for the RF link, and is required to demonstrate reliable telecommand uplink, telemetry downlink, forward error correction, retransmission-based reliability, and lightweight command authentication within a short simulated pass window.

2. System Architecture Overview

The subsystem follows the hybrid architecture permitted by Section 3.1 of the task brief: commercial LoRa RF modules (Semtech SX1276-class) provide the physical-layer radio front end, while all higher-layer protocol behaviour — framing, forward error correction, interleaving, retransmission control, authentication, and command dispatch — is implemented independently in firmware on STM32 microcontrollers. No black-box protocol handling is used; every field in every transmitted frame is defined and processed by the team's own code.

Figure 1 — End-to-End TT&C System Architecture



2.1 Space Segment

The space segment consists of a single STM32 'Black Pill' (STM32F4-class) microcontroller paired with a LoRa transceiver module. It hosts the full protocol stack (CRC, MAC, FEC, interleaving, replay protection) and interfaces with a simulated onboard control unit — an LED actuator representing a commandable spacecraft state, and a STATUS command path representing telemetry generation.

2.2 Ground Segment

The ground segment mirrors the space segment's protocol stack on a second STM32 node, and adds ground-only functions: multi-attempt automatic repeat request (ARQ) with bounded retries, ground-side Doppler pre/post-compensation, and a serial bridge (ESP32, USB-UART) to a Python-based operator GUI. The GUI provides command entry, live KPI display (packet error rate, ACK latency, Doppler shift, FEC-corrected bit count), and CSV-based telemetry archiving for post-pass analysis.

3. Link Budget Analysis

The link budget below is computed for the worst-case slant range at the lower bound of the target altitude band (500 km), using representative LoRa hardware parameters. It establishes the operating margin that justifies the selected data rate and spreading configuration. Antenna gains are treated as near-omnidirectional (dipole-class) at both ends, consistent with a low-cost CubeSat UHF whip and a ground-station monopole/Yagi combination.

Parameter	Value	Notes
Carrier frequency	433.0 MHz	UHF, license-exempt ISM sub-band region-dependent
TX power (both ends)	+17 dBm (50 mW)	LoRa module rated maximum, conducted
TX antenna gain	2 dBi	Dipole-class, CubeSat UHF whip
RX antenna gain	6 dBi	Ground Yagi / directional assumption
Slant range (worst case)	2,330 km	500 km altitude, 10 deg min elevation
Free-space path loss	≈ 143.7 dB	FSPL = $20\log_{10}(d) + 20\log_{10}(f) + 32.44$ (d in km, f in MHz)
Cable / connector losses	2 dB	Estimated, both ends combined
Receiver sensitivity	-137 dBm	LoRa, SF10/BW125kHz-class configuration
Received signal level	≈ -122.7 dB	TX EIRP - path loss - losses + RX gain
Link margin	≈ 14.3 dB	Received level minus RX sensitivity

A positive margin of approximately 14 dB at worst-case range provides headroom for fading, polarization mismatch, and the Doppler-induced frequency offset discussed in Section 4.4, and supports the selected data rate without requiring the highest (slowest, most robust) spreading factor. This margin should be re-validated experimentally in Phase 2 testing against the team's actual antenna hardware and measured TX power; the values above are datasheet-based design estimates, not measured results.

4. Modulation, Coding, and Doppler Strategy

4.1 Modulation Selection

LoRa's proprietary Chirp Spread Spectrum (CSS) modulation was selected over a discrete BPSK/GFSK implementation for three reasons: (1) CSS provides strong inherent robustness to Doppler shift and Doppler rate, since the chirp's frequency-time slope makes symbol detection tolerant of moderate carrier offsets without a dedicated carrier-recovery loop; (2) it provides processing gain suited to the power-constrained

CubeSat downlink without requiring a custom PHY built from discrete RF components, which was judged to carry unacceptable schedule risk for the competition timeline; and (3) it is supported by low-cost, flight-representative-form-factor modules compatible with the STM32 toolchain used throughout the stack.

The trade-off is architectural transparency at the physical layer: LoRa's internal preamble detection, symbol timing recovery, and carrier synchronization are vendor-implemented and not independently re-derived by the team. This is treated as a documented, justified simplification rather than an omission — see Section 6.

4.2 Forward Error Correction

Hamming(7,4) single-error-correcting block coding is applied independently to each 4-bit nibble of the 9-byte logical frame (18 codewords per frame), on top of LoRa's own CSS coding gain. Hamming(7,4) was selected over Reed-Solomon or a convolutional/Viterbi scheme for this competition cycle because it is deterministic, has fixed and predictable execution time suited to the STM32's real-time command-processing budget, and its correctness is straightforward to verify exhaustively (all 16 data patterns x all 7 single-bit-error positions), which was done during firmware validation. Reed-Solomon would provide stronger burst-error correction per symbol but at materially higher implementation and verification cost; this trade-off is revisited in Section 6.

4.3 Interleaving

A 3 x 6 block interleaver is applied to the 18 FEC codeword bytes before transmission (write row-major, transmit column-major). This ensures that a burst error spanning up to 3 consecutive wire bytes is distributed across 3 separate Hamming codewords rather than concentrating inside a single codeword, keeping each affected codeword within its single-bit correction capability. This behaviour was verified analytically prior to integration.

4.4 Doppler Compensation

Because the satellite node has no onboard GPS or orbit propagator, Doppler compensation is performed entirely on the ground segment, which is consistent with common small-satellite ground-station practice. The ground station applies a predicted frequency correction to both its transmit path (uplink pre-compensation) and its receive path (downlink post-compensation) using a simplified cosine model of Doppler shift over a pass:

$$f_{\text{shift}}(t) = f_{\text{max}} \cdot \cos(\pi \cdot t / T), \text{ for } 0 \leq t \leq T$$

where f_{max} is the configured worst-case shift (3.5 kHz, representative of a 433 MHz UHF LEO pass at 500-700 km) and T is the configured pass duration (600 s). This model approximates the characteristic Doppler curve of a LEO pass — large positive shift at acquisition of signal (AOS), zero at time of closest approach (TCA), large negative shift at loss of signal (LOS) — without requiring a full orbit propagator. It is explicitly a design simplification for the competition scope; a flight-representative implementation would replace this function with SGP4 propagation against a current two-line element (TLE) set and the ground station's known geodetic position.

5. Communication Protocol Selection

A custom CCSDS-inspired frame structure was selected over raw AX.25 framing. The logical (pre-FEC) frame is a fixed 9-byte structure: synchronization byte, source address, destination address, sequence number, command identifier, single-byte payload, message authentication code, and a 16-bit CRC. This was preferred over AX.25 because AX.25's variable-length, flag-delimited framing is designed for terrestrial packet radio and adds bit-stuffing overhead that complicates fixed-time embedded processing, whereas a fixed-length

CCSDS-style frame simplifies both the FEC/interleaving block sizing described in Section 4 and worst-case timing analysis on a resource-constrained MCU.

6. Trade-off Analysis Summary

Design Decision	Alternative(s) Considered	Basis for Selection
Hybrid architecture (commercial LoRa RF + custom protocol stack)	Fully custom discrete-RF baseband	Lower schedule and integration risk; still fully transparent above the PHY per Section 3.1 rules
LoRa CSS modulation	BPSK / GFSK on discrete RF	Inherent Doppler and low-SNR robustness; avoids building a custom PHY within the competition timeline
Hamming(7,4) FEC	Reed-Solomon, convolutional/Viterbi	Deterministic timing, exhaustively verifiable correctness, lower MCU resource cost
3x6 block interleaver	Larger/deeper interleaver, no interleaving	Matches the 18-byte FEC frame size exactly; deeper interleaving would add latency inconsistent with short pass windows
Ground-only Doppler compensation, cosine approximation	Onboard GPS + full SGP4 propagation	No onboard GPS in current hardware baseline; cosine model captures the dominant AOS/TCA/LOS shape at acceptable design complexity
Fixed 9-byte CCSDS-style logical frame	AX.25 variable-length framing	Simplifies fixed-block FEC sizing and worst-case execution timing on the STM32

7. Requirements Traceability Matrix

Requirement (Section)	Design Element	Status
Preamble / sync (4.1)	LoRa CSS preamble (vendor PHY)	Implemented (PHY), documented as vendor-provided
Doppler compensation (4.1, 4.4, 4.5)	Ground-side cosine model, TX/RX frequency correction	Implemented, simplified model
FEC (4.1, 4.5)	Hamming(7,4) per nibble	Implemented and unit-verified
Interleaving (4.1)	3x6 block interleaver	Implemented and verified
CRC (4.1)	CRC16-CCITT over 7-byte payload	Implemented
Packet framing, headers, sequence numbers (4.2)	9-byte logical frame with SEQ field	Implemented
Retransmission mechanism (4.2, 4.5)	Stop-and-wait, 3 retries, 800 ms timeout	Implemented
Command authentication (4.3, 4.5)	Keyed XOR MAC byte	Implemented, explicitly lightweight/demo-grade
Replay protection (4.3, 4.5)	8-entry sequence sliding window	Implemented
Doppler correction capability, ground (4.4)	GCS-side setFrequency() tracking	Implemented
Pass management / AOS-LOS (4.4)	Manual PASS_START trigger with timed window	Partially implemented — manual trigger, not autonomous detection
Data logging / telemetry archiving (4.4)	GUI CSV log of all KPI events	Implemented

8. Summary

The design satisfies the mandatory link-layer and command-protection requirements of Task IV in full, and satisfies the mandatory physical-layer requirements (FEC, interleaving, Doppler compensation) using documented, resource-appropriate simplifications suited to the competition timeline. The principal open item carried into Phase 2 testing is empirical validation of the link budget and Doppler model against measured hardware performance, and completion of autonomous (rather than manually triggered) pass management.